

FREDRICK ODUOR OTIENO

GIS & GEOSPATIAL DATA PROFESSIONAL

Kenya | fredoduor293@gmail.com

PROFESSIONAL PROFILE

Geospatial professional with a BSc in Geospatial Information Science with Information Technology and practical experience managing, processing, analysing and communicating spatial data. Experienced with ArcGIS Pro, QGIS and Global Mapper, with additional skills in Python, SQL/spatial databases, Google Earth Engine, remote sensing, field data collection and geospatial workflow automation. Detail-oriented and methodical, with experience maintaining cadastral, infrastructure, planning and environmental-related datasets, producing map outputs and supporting field and technical teams. Interested in applying GIS and spatial analysis to biodiversity, conservation and environmental decision-making.

CORE COMPETENCIES

- Geospatial data management and maintenance
- Spatial/tabular data quality assurance and validation
- GIS analysis, cartography and map production
- Data processing, documentation, archiving and file management
- ArcGIS Pro, QGIS and Global Mapper
- Remote sensing and environmental spatial analysis
- Python, GDAL, GeoPy and SQL/spatial databases
- GPS/GNSS and field geospatial data collection
- GIS workflow automation and GeoAI
- Technical reporting and stakeholder communication
- Microsoft Excel, Word and PowerPoint
- Independent working, prioritisation and deadline management

PROFESSIONAL EXPERIENCE

Geoinformation Intern - State Department for Housing and Urban Development, Kenya

Jan 2026 - Present | Bondo Affordable Housing Project

- Maintain and prepare geospatial datasets supporting project boundaries, utilities, infrastructure and cadastral information.
- Conduct topographical surveys, field data collection and setting-out activities, ensuring spatial information is captured accurately.
- Process, analyse and present spatial information for planning, zoning, environmental assessment support and project decision-making.
- Support quantitative analysis relating to housing allocation, project costs, demand and housing-market information.
- Coordinate geospatial information from field and project activities and communicate findings through maps, reports and other appropriate outputs.
- Work across multiple project requirements while maintaining accuracy, consistency and deadlines.

GIS Attaché - Ministry of Lands, Public Works, Housing and Urban Development, Kenya

May 2024 – Aug 2024

- Collected, processed and managed geospatial data for cadastral management, land records and urban planning.
- Prepared detailed maps supporting urban planning and county development activities.
- Supported management and updating of land-record and cadastral datasets, including mutations and subdivisions.
- Prepared survey plans using Civil 3D and participated in static GNSS survey and pre/post-processing.
- Participated in hydrological surveys and field data collection.

SELECTED GIS & REMOTE SENSING PROJECTS

Forest-cover and land-use change analysis - Used GIS and remote sensing to analyse land-use change and forest-cover dynamics.

Geospatial database and web application - Developed a spatial database for resource management at Maseno University and supported web-based visualisation.

Community mapping - Collected and organised geospatial information for community mapping around Maseno Township.

Crime spatial analysis - Applied GIS spatial analysis to examine crime patterns in North-Gem Ward, Siaya County.

TECHNICAL SKILLS

- GIS: ArcGIS Pro, QGIS, Global Mapper
- Remote sensing: Google Earth Engine, satellite imagery processing, raster analysis, land-use/land-cover analysis
- Programming/databases: Python, GDAL, GeoPy, SQL, spatial databases/geodatabases
- Data management: spatial and tabular datasets, data organisation, quality checking, documentation and file management
- Field/survey: GPS/GNSS, topographical surveys, cadastral data, static GNSS processing, Civil 3D
- Automation: GeoAI and Python-based GIS automation
- Office: Microsoft Excel, Word and PowerPoint

EDUCATION

BSc Geospatial Information Science with Information Technology - Maseno University, Kenya

2021 – 2025

Relevant areas: GIS, remote sensing, spatial databases, spatial analysis, cartography, surveying and geospatial information technology.

REFEREES

Mr. Joseph Okotto Okotto – University Lecturer

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